

Flatness of representation counts for prime sequences, I: the sub-peak spectrum and the second-order ripple

Kentaro Amauchi*

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Abstract

Let B be a finite set of odd primes and $r_B(m)$ the number of subsets of B summing to m . The companion paper reduces the asymptotic behaviour of the subset-sum landscape of B to a single quantity, the *flatness* ε_d , which measures the pointwise oscillation of r_B on a bounded window. This paper develops the Fourier-side input needed to bound ε_d : we determine the complete spectrum of sub-peaks of the characteristic product F_B , we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with m odd is exactly 0 whenever $m \in B$; for B a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \dots, 40$. Key steps are formally verified in Lean 4.

1 Introduction

1.1 The quantity to be bounded

For a finite set B of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \quad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta} d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant provided the *flatness*

$$\varepsilon_d = \max_{m, m' \in \text{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where B_d is the subsequence of elements exceeding $2d$ and Win_d is a bounded window about $T/2$. Bounding ε_d is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

*Independent researcher. amkn.sub03@gmail.com, <https://mathlab0911.github.io/>

1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a \in B} \cos(a\theta/2)$, so that $|F_B| = 2^b |G_B|$, where $b = |B|$, the natural per-element height at modulus q is

$$M(q) = \left(\prod_{\substack{1 \leq r \leq q \\ \gcd(r, q) = 1}} |\cos(\pi r/q)| \right)^{1/\varphi(q)}.$$

Proposition 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ (m odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction K_2 and phase drift ψ ; Section 5 treats the main arc and identifies the first correction to the central density, resolving an unexplained constant of the companion paper. Section 6 reports the numerical verification, Section 8 makes the finite list of remaining moduli effective, Section 9 treats the deep minor arcs, and Section 10 assembles the main theorem.

1.3 Relation to the literature

Three bodies of work are close enough to require comment.

Oscillations in partition counts. Murthy and Brack [MB26] give a semiclassical, periodic-orbit account of oscillations in partition counting functions; conceptually this is the same mechanism as ours, a few isolated saddle contributions producing a visible ripple. Their detailed oscillation analysis is for partitions into distinct *squares*, where the relevant orbits are indexed by Pythagorean triples, and their treatment of prime partitions (their §7) is asymptotic rather than oscillatory. The limit also differs: they let the integer being partitioned tend to infinity with the part set fixed, whereas we let the *number of parts* k tend to infinity. The dictionary is one line: their periodic orbits are our sub-peaks at rational $\theta = 2\pi j/q$. We would add that the question they leave open — why regular oscillations survive only for certain special sequences — has, in the subset-sum setting, the exact answer given by Lemma 3.1 and Corollary 3.2: an oscillation at modulus q survives precisely when q avoids the cyclotomic vanishing $q = 2m$ with $m \in B$, and the surviving heights are the $M(q)$ of Proposition 2.1.

Smoothness of partition functions. Bateman and Erdős [BE56] prove $p_A(n-1)/p_A(n) \rightarrow 1$ for a *fixed infinite* set A of parts with *repetitions allowed*. Our situation differs in both respects: our part set is a finite truncation that grows with k , and parts are used at most once, so the generating function is $\prod_{a \in B} (1 + x^a)$ rather than $\prod_{a \in A} (1 - x^a)^{-1}$. We do not see how to deduce the finite-truncation statement from theirs. The same remark applies to Roth and Szekeres [RS54] and to Vaughan [V08], both of which treat the fixed infinite set of all primes.

Landscapes of the number partitioning problem. Ferreira and Fontanari [FF98] compute the fraction of metastable states — that is, $\text{lm}/2^k$ — analytically for uniform random weights; Alyahya and Rowe [AR14] give a formula in terms of the coefficient of variation; Stadler, Hordijk and Fontanari [SHF03] analyse barrier trees. All of these average over *random* weights, whereas the companion paper and this one treat a *specific* sequence and an order-sensitive invariant.

Finally, the anti-concentration literature (Littlewood–Offord and its inverse forms) bounds the *maximum* of r_B ; what we need is a two-sided bound on the *ratio* at nearby points, which does not follow from a one-sided bound.

1.4 Formal verification

Statements marked `name` are machine-checked in Lean 4 against Mathlib; the repository builds with no `sorry` and no additional axioms. The canon currently consists of 97 theorems.

2 The sub-peak spectrum

The two statements of this section are classical: Proposition 2.1 is the resultant $\text{Res}(\Phi_q, x + 1)$ written out, and Lemma 2.2 is a standard exercise. We record both with proofs because we use the precise constants, and because the interpretation of $M(q)$ as the per-element height of a sub-peak of a subset-sum generating function seems not to have been made before.

Proposition 2.1 (Cyclotomic evaluation; classical). *For every integer $q \geq 3$,*

$$M(q) = \frac{1}{2} |\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q -th cyclotomic polynomial.

Proof. Put $\zeta = e^{2\pi i/q}$. For $1 \leq r \leq q - 1$ we have $|1 + \zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r,q)=1} (x - \zeta^r)$, evaluating at $x = -1$ gives $\prod_{\gcd(r,q)=1} |1 + \zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r,q)=1} |\cos(\pi r/q)| = 2^{-\varphi(q)} |\Phi_q(-1)|$. Take $\varphi(q)$ -th roots. $\square$

Lemma 2.2 (Classical values). *For $q \geq 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with p an odd prime and $j \geq 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \geq 2$; and $\Phi_q(-1) = 1$ otherwise.*

Proof. We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if n has at least two distinct prime factors; for odd $n \geq 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even m , $\Phi_{2m}(x) = \Phi_m(x^2)$. If q is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with m odd then $\Phi_q(-1) = \Phi_m(1)$, which is p when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with m even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when m is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. $\square$

Theorem 2.3 (Extremality of modulus six). $\sup_{q \geq 3} M(q) = \sqrt{3}/2$, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \leq 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.

Proof. By Proposition 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in j with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2} p^{1/(p^{j-1}(p-1))}$; the exponent decreases in j , so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2} p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \dots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2} 3^{1/6} \approx 0.6005$. Combining the three families gives the statement. $\square$

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

3 The vanishing family and the surviving spectrum

Lemma 3.1 (Exact vanishing). *Let $q = 2m$ with m odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

Proof. The factor of F_B indexed by m is $1 + e^{i\pi} = 0$. $\square$

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of m , so if B is a set of primes and $m > 1$ then B meets that class if and only if m itself lies in B . This gives a complete description.

Corollary 3.2 (Surviving spectrum). *Let B_d be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly, provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod{6}$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.
2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly, for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd a . This is the Fourier form of the modulus-4 obstruction of the companion paper.
3. $|G_{B_d}(2\pi/(2p))| = 0$ exactly, for every prime p with $2d < p \leq x$.
4. For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.
5. For $q = 2m$ with m odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2}3^{1/6} + o(1))^b$.
6. For q odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is 0 by parity.

Remark 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed d , only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes ≥ 5 ; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

4 The modulus-six ripple to second order

Throughout this section B is a set of b odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a \in B} a^2, \quad S_4 = \sum_{a \in B} a^4, \quad \Delta = \sum_{a \in B} \tau_a a, \quad \Delta_3 = \sum_{a \in B} \tau_a a^3,$$

where $\tau_a = +1$ if $a \equiv 1 \pmod{6}$ and $\tau_a = -1$ if $a \equiv 5 \pmod{6}$, and let c_1, c_5 be the corresponding class counts. Write $\text{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

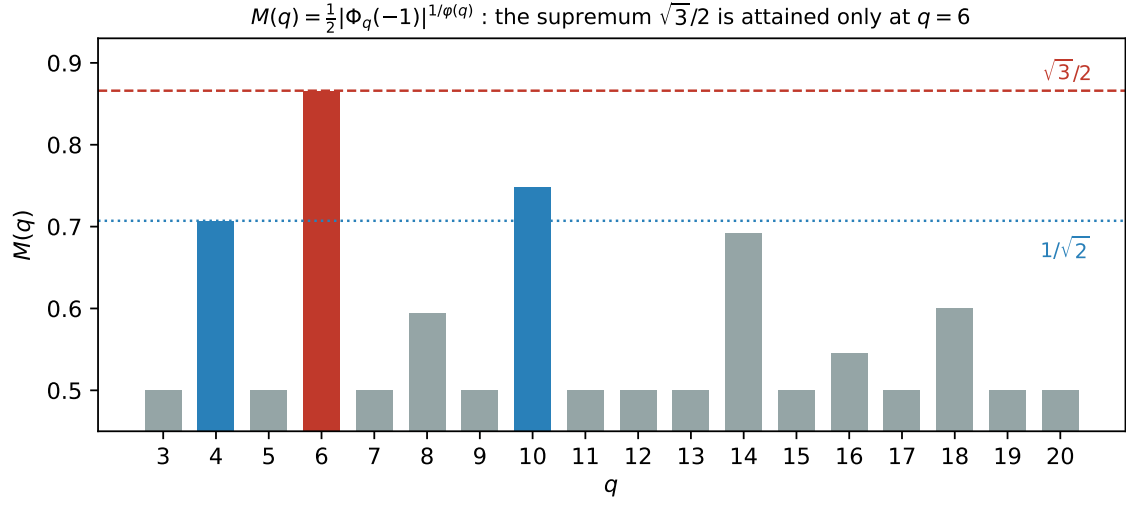


Figure 1: The per-element sub-peak height $M(q)$ (exact values, by Proposition 2.1). The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

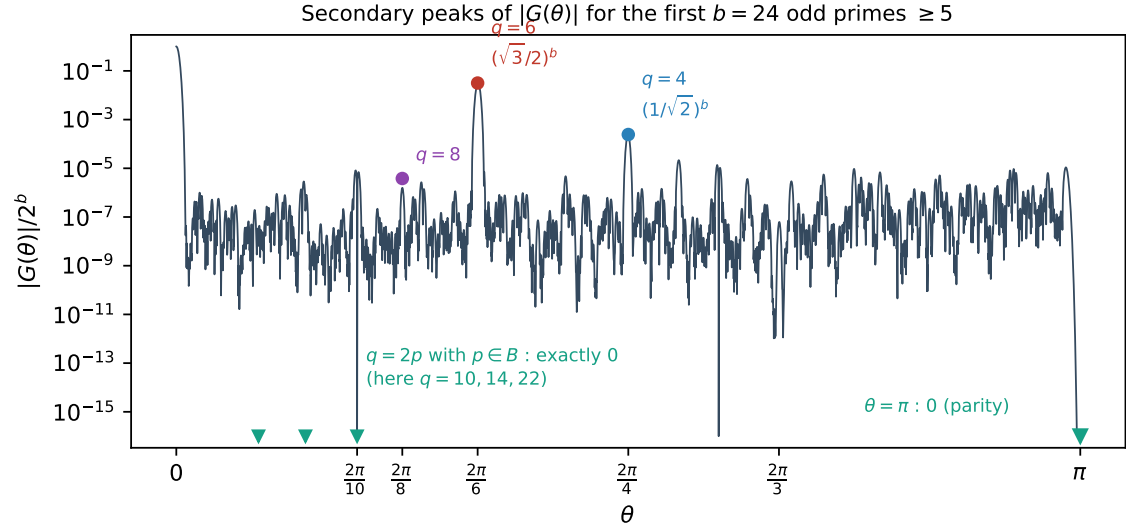


Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes ≥ 5 . The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1, proved); $\theta = \pi$ vanishes by parity. Computed exactly; no asymptotics enter this figure.

Lemma 4.1 (Local data at $\theta = \pi/3$). *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log |G_B(\pi/3 + t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

Moreover $\frac{d}{d\theta} \arg(1 + e^{ia\theta}) = a/2$ exactly, so the phase of F_B is linear along the arc and all curvature lives in $|G_B|$.

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_mul_div_six`.)

Proposition 4.2 (Second-order ripple formula). *For m in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\text{Main}(m)} - 1 = 2\left(\frac{\sqrt{3}}{2}\right)^{b+1} K_2 \cos\left(\frac{\pi}{6}(2m - (c_1 - c_5)) + \psi(\nu)\right) + E,$$

where

$$K_2 = \exp\left[\frac{\Delta^2}{8S_2} - \frac{3S_4}{8S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^3} - \frac{3S_4\Delta^2}{16S_2^3}\right], \quad \psi(\nu) = -\frac{\sqrt{3}}{2}\nu \frac{\Delta + \Delta_3/S_2}{S_2}.$$

Derivation. Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L + i\nu)^2/(2V_6) + C_3(L + i\nu)^3/(6V_6^3) + \dots$, the Gaussian-fluctuation factor $-\frac{1}{2}\log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \dots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives K_2 and ψ . The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b + 1$. $\square$

Remark 4.3 (The two branches). The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \pmod{6}$; the term $-\frac{3}{8}S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \geq 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \leq 47$ and $K_2 < 1$.

Remark 4.4 (Error policy). The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in Δ , of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

5 The main arc and the central density

The main arc contributes the smooth part of r_B . The leading term is the local central limit theorem; what matters for the companion paper's absolute formula is the *first correction* to it.

Proposition 5.1 (Central density with first correction). *Let B be a set of b odd integers, $V_0 = S_2/4$, and let m be at the centre $T/2$. Then*

$$r_B(m) = \frac{2^b}{\sqrt{2\pi V_0}} \left(1 - \frac{S_4}{4S_2^2} + O((S_4/S_2^2)^2)\right).$$

Derivation. Write $\log G_B(\theta) = \sum_{a \in B} \log \cos(a\theta/2) = -\frac{V_0}{2}\theta^2 - \frac{S_4}{192}\theta^4 + O(S_6\theta^6)$, using $\log \cos u = -u^2/2 - u^4/12 + O(u^6)$ and $u = a\theta/2$. Since G_B is even (Lean: `G_even`), all odd-order terms vanish identically, so the first correction is the quartic one. Integrating $e^{-V_0\theta^2/2}(1 - \frac{S_4}{192}\theta^4)$ over the main arc and using $\int \theta^4 e^{-V_0\theta^2/2} = 3\sqrt{2\pi}V_0^{-5/2}$ gives the stated factor $1 - \frac{S_4}{192} \cdot 3V_0^{-2} = 1 - \frac{S_4}{4S_2^2}$. $\square$

Remark 5.2 (The absolute formula of the companion paper). The companion paper predicts the number of strict local minima by $\text{lm} \approx \Gamma \cdot 2^b / \sqrt{2\pi V_0}$, i.e. by replacing the ground-state count by its Gaussian estimate. The measured ratio of the two sides sits systematically below 1 — about 0.97 — with no explanation offered there. Proposition 5.1 identifies the deficit as $S_4/(4S_2^2)$, which at $k = 20$ equals 0.0246. Section 6.4 verifies this. We warn against one confusion: the value 0.972 appearing here (the ratio lm/pred at $k = 20$) is *not* the 0.972 in the companion paper’s list of $Q(k)$, which is $Q(18)$. Those are different quantities and the numerical coincidence is accidental: Q is formed with the *measured* ground-state count, so it carries no Edgeworth correction, and it oscillates about 1 rather than sitting below it.

6 Numerical verification

All computations below are exact integer dynamic programming for r_B , followed by a discrete Fourier extraction; logs are included in the repository.

6.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

Δ_3^2 coefficient	$S_4\Delta^2$ term	branch A mean	branch B mean
45/24	no	0.00061	0.00736
45/24	yes	0.00060	0.00400
5/24	no	0.00073	0.00411
5/24	yes	0.00072	0.00073

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \dots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

On branch B the individual values of $\text{measured}/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$. The omitted contributions are dominated by the class-independent tail $O((S_4/S_2^2)^2)$, and S_4/S_2^2 decreases monotonically over the range, from 0.1111 at $k = 18$ to 0.0507 at $k = 40$; hence the residual is largest at the smallest k . That the Δ -dependent quantity Δ_3^2/S_2^3 is in fact largest at $k = 32$ ($2.70 \cdot 10^{-3}$, against $1.04 \cdot 10^{-3}$ at $k = 18$) without producing an excess residual there is corroborating evidence that the Δ -dependence is already absorbed into K_2 .

6.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \dots, 40$, only α is significant, at every window width tested: $t(\alpha)$ ranges from +4.26 to +5.16, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model $\text{measured} = c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths 24, 48, 96 and 192, with R^2 from 0.978 to

0.986; restricting to $|\Delta| \geq 17$ raises R^2 to 0.993. We conclude that ψ as stated is correct, with the deficit in c a finite-window effect.

Remark 6.1 (Extraction). The amplitude is extracted from a narrow window of fixed half-width 24, i.e. eight full periods of the modulus-6 ripple, which over $k = 18, \dots, 40$ is between 0.08 and 0.30 standard deviations; both requirements — enough periods, and narrower than σ — hold throughout. Wide windows corrupt the Gaussian reference. As a check, at $k = 18$ the ratio measured/ K_2 is 0.9988, 0.9974, 1.0034 and 1.0065 at half-widths 12, 16, 24 and 32, so the window choice moves the residual by less than 1%. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

6.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level d : a peak at $q = 2p$ is dark exactly when p is still present in B_d , and lights up as soon as the truncation removes p . All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

d	removed primes	$q = 6$	$q = 10$	$q = 14$	tallest
1	—	0	0	0	$q = 4$
2	3	$3.66 \cdot 10^{-2}$	0	0	$q = 6$
3	3, 5	$4.22 \cdot 10^{-2}$	$1.03 \cdot 10^{-3}$	0	$q = 6$
4	3, 5, 7	$4.88 \cdot 10^{-2}$	$1.75 \cdot 10^{-3}$	$4.33 \cdot 10^{-4}$	$q = 6$

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for A the first 24 odd primes. Entries equal to 0 are exact zeros.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every d , confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and 2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter. This is to be expected, since the error in the prime number theorem in arithmetic progressions decreases only logarithmically, so at $b = 21, \dots, 31$ the $o(1)$ in items (iv)–(v) of Corollary 3.2 has not yet become small.

6.4 The central density

We test Proposition 5.1 using only ground-state counts. Write $\lambda(B) = \deg(B)\sqrt{2\pi V_0}/2^b - 1$ for the measured relative deficit, where $\deg(B)$ is the number of subsets attaining the minimum of $|\sum S - \lfloor T/2 \rfloor|$. Proposition 5.1 predicts $\lambda \approx -S_4/(4S_2^2)$.

k	deg	measured λ	predicted	difference
16	394	−0.024764	−0.031338	+0.006573
18	1290	−0.025573	−0.027754	+0.002181
20	4344	−0.024036	−0.024629	+0.000594
24	51068	−0.021467	−0.020830	−0.000637
28	634568	−0.017663	−0.017574	−0.000090

Table 3: Central-density deficit for the first k odd primes. The agreement is within 0.5% for $k \geq 18$ and improves to 0.01% at $k = 28$.

Over an ensemble of 100 random odd sequences per size (drawn from the same range as the primes), the mean absolute difference is 0.0171 at $k = 16$, 0.0079 at $k = 18$ and 0.0038

at $k = 20$; the ensemble is noisier than the primes because $\deg$ is a small integer subject to arithmetic fluctuation.

The consequence for Remark 5.2 is direct. Writing $\rho = \text{lm}/(\Gamma \cdot 2^b/\sqrt{2\pi V_0})$ for the ratio in the absolute formula, the ensemble mean of ρ is 0.9462, 0.9621, 0.9672 and 0.9716 at $k = 12, 16, 18, 20$, while the mean of $\rho/(1 - S_4/(4S_2^2))$ is 0.9812, 0.9888, 0.9912 and 0.9933. The correction removes about three quarters of the deficit, and the share it removes grows with k . We therefore record the systematic part of the 0.97 as explained; the residual 0.7% at $k = 20$ is a separate and smaller effect, still decreasing.

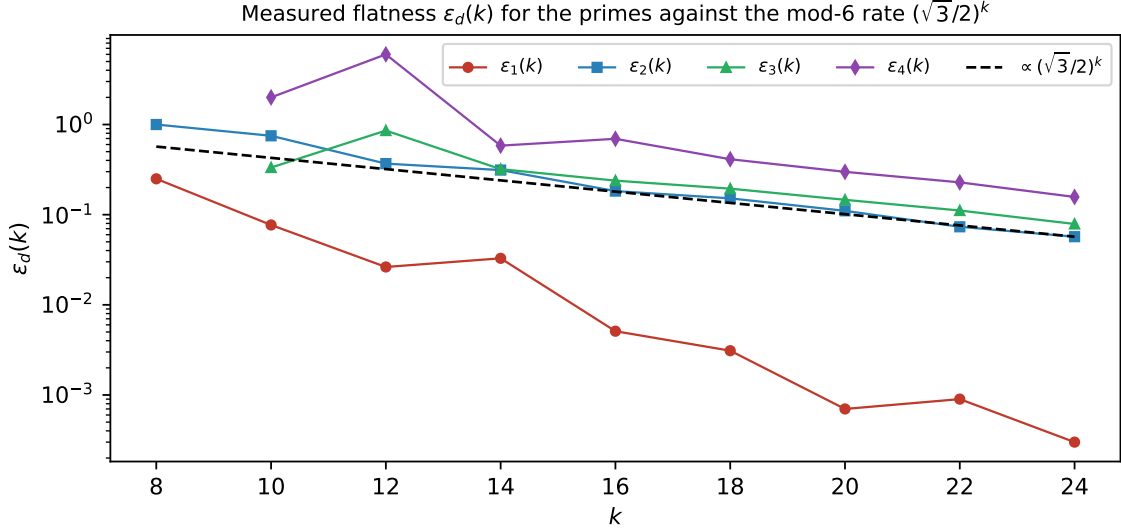


Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1). Measured by exact dynamic programming; the reference line is not a fit.

7 A free Gaussian envelope at every peak

The following is elementary but useful: every sub-peak carries an envelope as sharp as the main arc, with no arithmetic input at all.

Proposition 7.1 (Curvature bound). *Let B be any finite set of positive reals and $V_0 = \frac{1}{4} \sum_{a \in B} a^2$. On any interval where G_B has no zero,*

$$\frac{d^2}{d\theta^2} \log |G_B(\theta)| = - \sum_{a \in B} \frac{a^2}{4} \sec^2\left(\frac{a\theta}{2}\right) \leq -V_0,$$

because $\sec^2 \geq 1$. Consequently $\log |G_B|$ is concave there with curvature at least V_0 , and if θ^ is an interior local maximum then*

$$|G_B(\theta)| \leq |G_B(\theta^*)| e^{-V_0(\theta - \theta^*)^2/2}$$

throughout that interval.

(Lean: `one_le_one_div_cos_sq`, `term_curvature_ge` and `curvature_ge_V0` formalise the inequality $V_0 \leq \sum (a^2/4) \sec^2(a\theta/2)$, and `deriv_le_of_second_deriv_nonpos` together with

`le_of_critical_of_second_deriv_nonpos` formalise the analytic step: if $g'' \leq 0$ and $g'(t) = 0$ then t is a maximum of g . Substituting $g = f + c(\cdot - t)^2/2$ is then mechanical.)

Two caveats matter in practice, and both are visible numerically.

Remark 7.2 (The centre is not $2\pi/q$). The envelope must be anchored at the true local maximum, which is displaced from $2\pi/q$ by $-h'(0)/h''(0)$ with $h = \log |G_B|$; by Lemma 4.1 this is $\asymp \Delta/S_2$ at $q = 6$. For the first 24 odd primes truncated at $d = 2$ the measured displacements are $+1.316 \cdot 10^{-5}$ ($q = 6$), $+3.795 \cdot 10^{-4}$ ($q = 4$) and $-2.097 \cdot 10^{-3}$ ($q = 8$), against predicted $+1.316 \cdot 10^{-5}$, $+3.800 \cdot 10^{-4}$ and $-2.021 \cdot 10^{-3}$. Anchoring at $2\pi/q$ instead makes the envelope fail, by $7.6 \cdot 10^{-6}$ at $q = 6$ and $4.7 \cdot 10^{-3}$ at $q = 4$ in the same data.

Remark 7.3 (Validity radius). The bound holds only inside the zero-free interval, whose half-width is $\approx \pi/(2 \max B)$. Measured in units of the envelope scale $V_0^{-1/2} = 2/\sqrt{S_2}$, this is

$$\frac{\pi}{2 \max B} \cdot \frac{\sqrt{S_2}}{2} \approx \frac{\pi}{4} \sqrt{k/3} \approx 0.45\sqrt{k},$$

so the envelope covers a growing number of standard deviations: 1.72 at $k = 16$, 2.08 at $k = 24$, 2.71 at $k = 40$ and 2.92 at $k = 48$, against the prediction 1.80, 2.21, 2.85, 3.12.

8 Effective constants for the surviving peaks

Items (iv) and (v) of Corollary 3.2 are the only ones that use equidistribution, and only for a finite list of moduli. We make them effective.

Write $n_r = \#\{p \in B_d : p \equiv r \pmod{q}\}$ for $\gcd(r, q) = 1$, and $e_r = n_r - b/\varphi(q)$. Since $\log |G_{B_d}(2\pi j/q)| = \sum_r n_r \log |\cos(\pi r j/q)|$ and $\sum_r e_r = O(1)$,

$$|G_{B_d}(2\pi j/q)| \leq M(q)^b \exp\left(L_q \sum_r |e_r|\right), \quad L_q = \max_{\gcd(r, q)=1} |\log |\cos(\pi r/q)||.$$

Bennett, Martin, O'Bryant and Rechnitzer [BMOR18] supply two explicit inputs which between them cover all x : their Theorem 1.3 gives $|\pi(x; q, a) - \text{Li}(x)/\varphi(q)| < c_\pi(q) x/(\log x)^2$ for $x \geq x_\pi(q)$, with $c_\pi(q) \leq 1/840$ and $x_\pi(q) \leq 8 \cdot 10^9$ for $3 \leq q \leq 10^4$; and their Theorem 1.9 gives $\max_{y \leq x} |\pi(y; q, a) - \text{Li}(y)/\varphi(q)| \leq 2.734\sqrt{x}/\log x$ for $x \leq x_2(q)$, where $x_2(q) = 10^{13}$ for $5 < q \leq 100$. The ranges overlap, so there is no gap.

Proposition 8.1 (Effective form of Corollary 3.2(iv)–(v)). *For each $\eta > 0$ and each modulus q in the finite list there is an explicit $k_0(q, \eta)$ such that $|G_{B_d}(2\pi j/q)| \leq (M(q) + \eta)^b$ for all $k \geq k_0(q, \eta)$. With the constants of [BMOR18] one may take*

q	4	6	8	10	14	18
$k_0(q, 0.10)$	108	34	1694	856	3547	4411
$k_0(q, 0.05)$	343	104	5596	2839	11914	14723
$k_0(q, 0.01)$	5984	1737	103960	52658	226200	278350

Remark 8.2. The $\sqrt{x}$ -shaped bound of [BMOR18, Thm. 1.9] is what makes these thresholds small. Using only [BMOR18, Thm. 1.3] one is forced to $x \geq x_\pi(q) \approx 10^9$, i.e. $k_0 \approx 4 \cdot 10^8$; the two theorems must be combined. It is worth recording that using only the first of the two costs a factor of 10^4 in the threshold.

Regarding the range of validity we are explicit in three steps. (a) For $k \leq 44$ the peak heights are certified by direct computation: $|G_{B_d}(2\pi j/q)|$ is a product of b cosines and costs $O(k)$ arithmetic operations, so it is simply evaluated (Section 6). (b) For $k \geq k_0(q, \eta)$ the bound is a theorem, by Proposition 8.1. (c) The gap between the two is not needed for any asymptotic statement, and in any case it can be closed by computation: for each of

the finitely many q involved, evaluating $|G_{B_d}(2\pi j/q)|$ for all k up to any desired bound is an $O(k)$ computation per point. Note also that $q = 4$ and $q = 6$ appear in the table only for comparison: for those two moduli Corollary 3.2(i)–(ii) are exact identities and no equidistribution input is needed at all.

9 The deep minor arcs

This section is a design, not a proof; we state it because it isolates exactly what is missing and because one structural observation makes the shape of the argument clear.

9.1 The product is a superposition of prime exponential sums

Expanding $\log |2 \cos \pi u| = \sum_{n \geq 1} (-1)^{n+1} \cos(2\pi n u)/n$ termwise,

$$\log |G_B(\theta)| = -b \log 2 + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \operatorname{Re} \sum_{a \in B} e\left(\frac{na\theta}{2\pi}\right). \quad (1)$$

The inner sum is a classical exponential sum over primes. Thus the object we must bound on the deep minor arcs is not exotic: it is a weighted superposition of the sums $\sum_{p \leq x} e(\alpha p)$ that Vinogradov’s method was built for, with $\alpha = n\theta/2\pi$. The only delicacy is that the coefficients $1/n$ decay slowly, and that the series diverges at the zeros of $\cos$, so a truncation is needed near those points.

9.2 How the circle is divided, and why nothing falls between

Before the argument we fix the dissection, because the choice of the truncation parameter N_0 decides whether it is exhaustive. Let $A \geq 2$ be the exponent in the Siegel–Walfisz theorem, put $A' = A + 1$, and set

$$\delta_0 = \frac{1}{\log b}, \quad N_0 = (\log N)^{A'}, \quad Q_2 = (\log N)^{A+A'}.$$

Tier	range of q	tool	located in
exact peaks	$q \in \{4, 6\} \cup \{2p : p \in B\}$	cyclotomic identities	§3
T_2	remaining $q \leq Q_2$	PNT in AP + envelope	§8
T_3	deep minor, see (ii) below	Vaughan + Selberg + Abel	this section

In tier T_2 the equidistribution of B in the residue classes mod q is what produces the constant $M(q)$. Asymptotically this is the Siegel–Walfisz theorem, which is what fixes A ; effectively, for the q that occur at any accessible k , it is the explicit bounds of [BMOR18] used in Proposition 8.1. Only the ineffectivity of the Siegel zero prevents the two from being the same statement, and that is the sole reason Section 8 carries a threshold rather than a constant.

Lemma 9.1 (Covering completeness). *Every θ satisfies at least one of*

- (i) $|\theta - 2\pi j/q| \leq 1/(q\tau)$ for some $q \leq Q_2$ and $(j, q) = 1$;
- (ii) for every $n \leq N_0$ the reduced denominator q_n of $n\theta/2\pi$ exceeds $(\log N)^A$.

Proof. Suppose (ii) fails: some $n \leq N_0$ has $q_n \leq (\log N)^A$, so $|n\theta/2\pi - j'/q_n| \leq 1/(q_n\tau)$ for some j' . Dividing by n exhibits $\theta/2\pi$ within $1/(nq_n\tau)$ of $j'/(nq_n)$, and $nq_n \leq N_0(\log N)^A = Q_2$. Reducing the fraction only decreases the denominator, which gives (i). $\square$

Remark 9.2. With the earlier choice $N_0 = b^{1/4}$ the same computation gives $nq_n \leq b^{1/4}(\log N)^A$, and the band $(\log N)^A < q \leq b^{1/4}(\log N)^A$ then belongs to no tier: too large for the explicit bounds, too small to be deep minor. Replacing $b^{1/4}$ by a fixed power of $\log N$ closes it, and costs nothing — see Remark 9.7, where the change in fact improves every measured quantity that matters.

9.3 The intended argument

Fix θ of type (ii); the θ of type (i) are covered by Corollary 3.2 and Proposition 8.1. Split $B = B_{\text{good}} \cup B_{\text{bad}}$, where p is bad when $\|p\theta/2\pi - \frac{1}{2}\| < \delta_0$.

1. *Harmonics stay minor.* This is exactly condition (ii): for every $n \leq N_0$ the reduced denominator of $n\theta/2\pi$ exceeds $(\log N)^A$, so every harmonic appearing below lies on a minor arc. This is the only place where the value of N_0 enters the dissection, and it is why N_0 must be a power of $\log N$ rather than of b (Remark 9.2).
2. *The exponential sums.* Take the Dirichlet parameter $\tau = N(\log N)^{-A}$, so that every denominator arising above satisfies $q_n \leq \tau$ automatically; combined with step 1,

$$(\log N)^A < q_n \leq N(\log N)^{-A},$$

which is precisely the range on which Vaughan's identity is effective. No other range occurs: the tiers below $(\log N)^A$ are not lost but are handled in §3 and §8, where the cyclotomic constants $M(q)$ give the bound directly and no exponential sum is needed at all. On this range Vaughan's identity gives $S_B(\alpha) = \sum_{p \in B} e(p\alpha) \ll (Nq_n^{-1/2} + N^{4/5} + (Nq_n)^{1/2})(\log N)^4$, hence $S_B \ll N(\log N)^{4-A/2}$. Since $\pi(N) \asymp N/\log N$, this is $o(\pi(N))$ as soon as $A > 10$; A is ours to choose, and Siegel–Walfisz holds for every fixed A , so we fix $A = 11$ once and for all. A log-free version valid on a wider range of q is available in [Sr25]. Summing the head, $\sum_{n \leq N_0} |S_B(n\theta)|/n \leq (\log N_0) \max_n |S_B(n\theta)|$, gives $o(b)$; note that $\log N_0 = A' \log \log N$, so enlarging N_0 from $b^{1/4}$ to $(\log N)^{A'}$ costs the head only a bounded factor.

3. *Bad points.* Each bad p contributes at most $\log 2$ to (1), so it suffices that $|B_{\text{bad}}| = o(b)$. This needs no input beyond step 2. Lemma 12.1 of the appendix — the one-interval form of the Erdős–Turán inequality, proved there from the Fejér kernel so that no external reference is incurred — applied to the interval $\|x - \frac{1}{2}\| < \delta_0$ with degree N_0 , gives

$$|B_{\text{bad}}| \leq 4b\left(\delta_0 + \frac{1}{N_0}\right) + 4 \sum_{n \leq N_0} c_n |S_B(n\theta)|, \quad c_n = \min\left(2\delta_0 + \frac{2}{N_0}, \frac{1}{\pi n}\right),$$

and the sum on the right is the *same* head already shown to be $o(b)$ in step 2. Since $\delta_0 \rightarrow 0$ and $N_0 \rightarrow \infty$, the whole expression is $o(b)$.

We stress the point, because it removes a dependency rather than adding one: no external discrepancy theorem for $\{p\theta\}$ is required. The literature on αp modulo one — Vinogradov, Vaughan, Harman, Matomäki — proves *small-value* statements, that some prime $p \leq N$ makes $\|\alpha p - \beta\|$ small; that is a statement about the extreme of the sequence and does not by itself bound its discrepancy. The route above uses only the exponential sums we already have.

4. *Good points.* For p good the series (1) is truncated at N_0 ; since $\|p\theta/2\pi - \frac{1}{2}\| \geq \delta_0$, summation by parts against the Dirichlet kernel bounds the tail by $C/(N_0\delta_0) =$

$C \log b (\log N)^{-A'} = o(1)$. Here the new choice of N_0 is a large gain and not merely a repair: with $N_0 = b^{1/4}$ the same bound was $C \log b \cdot b^{-1/4}$, which stays above 1 until $b \approx 10^6$, whereas $(\log N)^{A'}$ makes it small immediately (Remark 9.7).

Together these give $\log |G_B(\theta)| \leq -(b - o(b)) \log 2 + o(b)$, that is $|G_B(\theta)| \leq (1/2 + o(1))^b$ on the deep minor arcs.

Remark 9.3 (What this settles about the technique needed). Only a one-sided bound together with an exceptional set of size $o(b)$ is required. The precise asymptotic machinery of [DRZZ25] — their strange and pseudo-differentiable functions — is therefore not needed here, and we cite them for context only. The question was nevertheless the right one: the non-smoothness of $\log |2 \cos|$ is exactly where the elementary excision of step 4 does its work.

Remark 9.4 (An alternative route). One can instead avoid (1) altogether and argue through the full discrepancy $D_B(\theta)$ of $\{p\theta/2\pi\}_{p \in B}$: bound it by Erdős–Turán from the same exponential sums, then apply Koksma’s inequality to the truncated function $\max(\log |\cos \pi x|, \log \delta)$. The two routes give the same conclusion. We prefer the one above for two reasons: (1) is an exact identity, and step 3 needs the Erdős–Turán machinery only for a single interval, where it is elementary, rather than uniformly over all intervals.

Remark 9.5 (Where the effective constants are, and are not). Effectivity in this paper is confined to three places: the peak heights at $q = 4, 6$, which are exact identities (Corollary 3.2(i)–(ii)); the remaining finite list of moduli, effective with $k_0 \leq 1.5 \cdot 10^4$ (Proposition 8.1); and the Gaussian envelopes, which need no arithmetic input at all (Proposition 7.1). This section is asymptotic; see Remark 9.6.

Remark 9.6 (On explicit constants). One might hope to make step 3 effective using Helfgott’s explicit minor-arc bounds [Hel12]. His Main Theorem is stated for $x \geq x_0 = 2.16 \cdot 10^{20}$, which corresponds to $k = \pi(x_0) \approx 4.7 \cdot 10^{18}$. That is far outside any range we could combine with the direct computations of Section 6, so we do not claim an effective version of this section. Effectivity in this paper is confined to Proposition 8.1 (thresholds $\approx 10^4$) and to Proposition 7.1, which needs no arithmetic input at all.

Remark 9.7 (Numerical check of the budget). The parameters are asymptotic and are meaningless at $b = 23$, so we tested the steps above at b up to 1999, taking the worst of six sampled deep-minor θ at each size. Each evaluation is $O(bN_0)$. Writing tail for the step-4 bound $1/(N_0\delta_0)$:

b	99	299	999	1999
$ B_{\text{bad}} /b$ (measured)	0.485	0.405	0.324	0.282
step-3 bound/ b , $N_0 = (\log N)^3$	1.079	0.910	0.614	0.459
$b^{-1} \sum_{n \leq N_0} S_B(n\theta) /n$	0.697	0.640	0.415	0.292
tail, $N_0 = b^{1/4}$ (old)	1.532	1.425	1.381	1.267
tail, $N_0 = (\log N)^3$ (new)	0.047	0.031	0.021	0.017

The bound of step 3 holds in every case. The change of N_0 is the decisive one: with $N_0 = b^{1/4}$ the tail bound never drops below 1 in any accessible range — it would need $b \approx 10^6$ — so that step was vacuous in practice, whereas with $N_0 = (\log N)^3$ it is already below 0.05 at $b = 99$. The price is a larger head, since more harmonics are summed; the head is the slower of the two now, and the step-3 bound only becomes nontrivial (that is, smaller than 1) from $b \approx 300$.

None of this is close to binding. The conclusion needs only $\frac{1}{2}e^{o(1)} < \sqrt{3}/2$, i.e. the $o(1)$ to stay below $\log \sqrt{3} = 0.549$, and the quantity it controls, $-b^{-1} \log |G_B(\theta)| - \log 2$, was measured at 0.123 for $b = 99$ and 0.024 for $b = 1999$.

Remark 9.8 (What the peaks actually look like). As a check on the dissection we located every local maximum of $|G_B|$ on $[0, \pi]$ by direct evaluation on a grid of $6 \cdot 10^6$ points, for $k = 40, 60, 100, 140$. Every one of the twelve largest lies within the envelope radius of a rational $2\pi j/q$ with $q \leq 26$, and the ordering by height follows $M(q)$ as predicted, with one instructive exception: at $q = 10$, where $5 \in B$ forces $G_B(2\pi/10) = 0$ exactly, the vanishing cuts a narrow notch and the flanking maxima sit at $b^{-1} \log |G_B| = -0.355$ at $b = 99$, rising towards $\log M(10) = -0.291$ like $-\frac{3}{2}(\log b)/b$, which is the expected width of the notch. Excluding the central peak and a disc of the envelope radius about each rational with $q \leq 60$, the largest remaining value of $b^{-1} \log |G_B|$ is -0.305 at $b = 99$ and -0.305 at $b = 139$, against $\log(\sqrt{3}/2) = -0.144$: a margin of 0.16 per element, stable in k .

10 The main theorem

Nothing new is proved in this section; it is the inventory that assembles the previous ones.

Theorem 10.1 (Asymptotic flatness, conditional). Assume (a) the deep-minor bound of Section 9, i.e. that $|G_B(\theta)| \leq (1/2 + o(1))^b$ for θ of type (ii) of Lemma 9.1 — that section establishes it at the level of a sketch — and (b) that the envelopes of Proposition 7.1 reach as far as the Dirichlet neighbourhoods of type (i), which is Problem 11.3. Fix $d \geq 2$ and let $n = \lfloor T/2 \rfloor$. With $B = B_d(k)$, $b = |B|$, as $k \rightarrow \infty$,

$$\varepsilon_d(k) = 4 \left(\frac{\sqrt{3}}{2} \right)^{b+1} K_2(k) \kappa_d(k) (1 + o(1)) + O\left(\frac{w_d^2}{S_2}\right),$$

where K_2 is the amplitude correction of Proposition 4.2, $\kappa_d \in [0, 1]$ is the explicit trigonometric window factor — half the range of $\cos((\pi/6)(2m - (c_1 - c_5)) + \psi)$ over $m \in \text{Win}_d$ — and w_d is the window width. In particular $\varepsilon_d(k) \rightarrow 0$ at the geometric rate $\sqrt{3}/2$ per element.

Remark 10.2. The hypotheses are not decorative: they are the two statements of this paper that are not proved, and we prefer to carry them explicitly rather than bury them. Hypothesis (a) disappears once the programme of Section 9 is written out in full; hypothesis (b) is a bounded amount of work on the envelope and is supported by the measurements of Remark 9.8.

Corollary 10.3 (conditional, as above). Combining Theorem 10.1 with the sandwich theorem of the companion paper and the identity $W_D = \Gamma + (2D + 1)2^{-k}$,

$$\frac{\text{lm}_P(n)}{\deg_P(n)} \longrightarrow \Gamma(P) = 5.34920 \dots$$

for the odd primes.

Outline. The bookkeeping is as follows. (1) By Proposition 5.1, $r_B(m) = \text{Main}(m)(1 + E_0(m))$ with the correction $1 - S_4/(4S_2^2)$; over the bounded window the smooth ratio varies by $O(w_d^2/S_2)$. (2) By Proposition 4.2, the two exact peaks contribute $2(\sqrt{3}/2)^{b+1} K_2 \cos(\cdot + \psi)$ and $2(1/\sqrt{2})^{b+1} K_4 \cos(\cdot)$; the second is smaller by a factor $(0.8165)^b$ and is absorbed into $1 + o(1)$. (3) By Lemma 3.1 every peak at $q = 2p$ with $p \in B$ vanishes, and by Corollary 3.2 and Proposition 8.1 the surviving finite list has heights at most $(m_d + \eta)^b$ with $m_d \leq 5^{1/4}/2$. (4) By Proposition 7.1 each peak carries a Gaussian envelope of curvature V_0 centred at its true maximum, with reach $\Theta(\sqrt{k})$ standard deviations, so the arcs tile. (5) Off all envelopes, Section 9 gives $|G_B| \leq (1/2 + o(1))^b$. (6) Integrating over each region and dividing by Main

gives the inventory

region	bound, relative to Main	measured, $k = 32$
central	1	0.985
$q = 6$	$C(\sqrt{3}/2)^b$	$1.03 \cdot 10^{-2}$
$q = 4$	$C(1/\sqrt{2})^b$	$1.48 \cdot 10^{-5}$
other $q \leq Q_2$	$CQ_2(m_d + o(1))^b$	$4.06 \cdot 10^{-7}$
deep minor	$CQ_2(m_d + o(1))^b + C(1/2 + o(1))^b$	$3.66 \cdot 10^{-6}$

with $m_d \leq M(18) = 0.6005$ for $d = 2$. The last row deserves a word. It is tempting to bound the deep-minor region by its pointwise bound $(1/2 + o(1))^b$ times the measure of the region, but that is wrong by several orders of magnitude — at $k = 32$ it would give $1.2 \cdot 10^{-7}$ against a measured $3.7 \cdot 10^{-6}$ — because the integral there is dominated not by the deepest points but by the many shallow peaks the region also contains, each of which carries a width. Counting those peaks with their widths, at height m_d^b , reproduces the measurement to within a bounded factor. The conclusion is unaffected: what matters is that every row after the second is $o((\sqrt{3}/2)^b)$, and the measured ratios of rows 3–5 to row 2 fall from 1.3% at $k = 24$ to 0.04% at $k = 32$. Taking the maximum and minimum over the bounded window then gives the displayed formula. $\square$

11 Honest scope, and what remains

We set out plainly which parts of this paper are proved and which are not.

- *Proved on paper:* Proposition 2.1 and Lemma 2.2 (classical), Theorem 2.3, Lemma 3.1, Corollary 3.2(i)–(iii) and (vi), Proposition 7.1, Lemma 9.1, Lemma 12.1, Theorem 13.1.
- *Derivations verified numerically to stated precision:* Proposition 4.2 (K_2 and ψ , to 0.1% over $k = 18, \dots, 40$) and Proposition 5.1 (the Edgeworth term).
- *Effective:* Proposition 8.1 ($k_0 \leq 1.5 \cdot 10^4$) and Corollary 3.2(i)–(ii), which are exact.
- *Sketch level, to be written to referee standard:* Section 9 (the Vinogradov input of step 2 and the excision of step 4), Proposition 13.3, and the $o(1)$ bookkeeping in item (6) of the outline above. Theorem 10.1 and Corollary 10.3 inherit this status.
- *A gap we have found and not closed.* Lemma 9.1 sorts every θ by *denominator*, but the two mechanisms it hands θ to are matched to different *radii*: the Dirichlet neighbourhood in case (i) has radius $1/(q\tau)$, while the envelope of Proposition 7.1 is certified only out to $\approx 0.45\sqrt{k}\sigma$, smaller by a power of $\log N$. Nothing in this paper bounds $|G_B|$ on the annulus between the two. The measurements of Remark 9.8 say the annulus is harmless — the largest value anywhere outside the envelopes is 0.16 per element below $\log(\sqrt{3}/2)$, stably in k — but that is a measurement, not a proof. The natural repair is to widen the certified envelope by applying the curvature bound to a sub-product $\{a \in B : a \leq A_1\}$, whose zeros are spaced $2\pi/A_1$ apart; since $|G_B|$ is at most the sub-product, this costs only a smaller V_0 . We record it as Problem 11.3.
- *Formal verification:* the 97 Lean theorems cover the combinatorial core — the identities of both papers, the curvature bound, and the vanishing lemma. The analytic sections are not formalised and we do not claim that they are.

The remaining work is therefore of three kinds, and none of it requires a new idea: writing Section 9 and Proposition 13.3 out in full, discharging the $o(1)$ inventory, and the usual polishing. We state the three steps as problems for the record.

Problem 11.1 (Main arc). Make the local central limit theorem for r_B on the main arc effective, with an error term of the shape C/S_2 arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in b .

Problem 11.2 (Minor arcs). Carry out the programme of Section 9. The exponential sums are the classical ones by (1); the non-classical part is the excision near the zeros of $\cos$. Dong, Robles, Zaharescu and Zeindler [DRZZ25] handle non-principal major arcs for prime partitions using what they call strange and pseudo-differentiable functions, and their treatment of non-smooth behaviour looks like the right tool for that excision, although their limit ($n \rightarrow \infty$ with the part set fixed) is not ours.

Problem 11.3 (The annulus). Extend the certified reach of the envelope of Proposition 7.1 from $\Theta(\sqrt{k})$ standard deviations to the full Dirichlet neighbourhood $|\theta - 2\pi j/q| \leq 1/(q\tau)$, so that cases (i) and (ii) of Lemma 9.1 meet. Applying the curvature bound to the sub-product over $\{a \in B : a \leq A_1\}$ should suffice: its factors have zeros spaced $2\pi/A_1$ apart, and $|G_B|$ never exceeds it. One must check that the resulting $V_0(A_1) = \sum_{a \leq A_1} a^2/4$ is still large enough at the radius required.

Problem 11.4 (Assembly). Combine these to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \rightarrow 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for ε_d can be extracted from the aggregate residue-class spread. The route to Problem 11.4 must go through the upper bounds above.

12 Appendix: a self-contained one-interval Erdős–Turán bound

Step 3 of Section 9 needs to count the points of a finite family that land in one short interval. The Erdős–Turán inequality does this uniformly over all intervals, at the price of the Beurling–Selberg extremal function. For a single interval one can construct the majorant from the Fejér kernel in half a page, which is what we do, so that the section incurs no external reference.

Let $F_M(t) = \sum_{|n| \leq M} (1 - \frac{|n|}{M+1}) e(nt) = \frac{1}{M+1} \left(\frac{\sin \pi(M+1)t}{\sin \pi t} \right)^2$ be the Fejér kernel: a nonnegative trigonometric polynomial of degree M with $\int_0^1 F_M = 1$.

Lemma 12.1. *Let $x_1, \dots, x_b \in \mathbb{R}/\mathbb{Z}$, let I be an interval of length 2δ , and let $M \geq 1$. Put $S(n) = \sum_{j \leq b} e(nx_j)$. Then*

$$\#\{j : x_j \in I\} \leq 4b \left(\delta + \frac{1}{M+1} \right) + 4 \sum_{n=1}^M \min \left(2\delta + \frac{2}{M+1}, \frac{1}{\pi n} \right) |S(n)|.$$

Proof. Write $w = 1/(M+1)$. For $w \leq |t| \leq \frac{1}{2}$ one has $\sin \pi t \geq 2t$, so $F_M(t) \leq (4(M+1)t^2)^{-1}$ and hence $\int_{w \leq |t| \leq 1/2} F_M \leq \frac{1}{2(M+1)w} = \frac{1}{2}$; consequently $\int_{|t| < w} F_M \geq \frac{1}{2}$.

Let J be the interval with the same centre as I and length $2\delta + 2w$, and set $P = 2(\mathbf{1}_J * F_M)$. Then P is a trigonometric polynomial of degree at most M . If $x \in I$ then $x - J \supseteq (-w, w)$, so $(\mathbf{1}_J * F_M)(x) = \int_{x-J} F_M \geq \frac{1}{2}$ and $P(x) \geq 1$; thus $P \geq \mathbf{1}_I$ pointwise. Its coefficients are $\widehat{P}(n) = 2\widehat{\mathbf{1}}_J(n)(1 - \frac{|n|}{M+1})$, and $|\widehat{\mathbf{1}}_J(n)| = |\frac{\sin(\pi n|J|)}{\pi n}| \leq \min(|J|, \frac{1}{\pi|n|})$, while $\widehat{P}(0) = 2|J| = 4\delta + 4w$. Therefore

$$\#\{j : x_j \in I\} \leq \sum_{j \leq b} P(x_j) = \sum_{|n| \leq M} \widehat{P}(n) \overline{S(n)} \leq b\widehat{P}(0) + 2 \sum_{n=1}^M |\widehat{P}(n)| |S(n)|,$$

which is the stated bound. $\square$

Remark 12.2. The constant 4 is wasteful — the Beurling–Selberg majorant replaces it by 1 — but Section 9 only needs $o(b)$, and the bound is applied with $\delta = \delta_0 \rightarrow 0$ and $M = N_0 \rightarrow \infty$, so absolute constants are immaterial. The numerical values of the resulting bound are tabulated in Remark 9.7.

13 Appendix: the companion paper’s Problem on random comparison

The companion paper asks whether $\text{lm}_{P_k}/\text{lm}_{R_k} \rightarrow 0$, where P_k is the first k odd primes and R_k a random odd sequence of the same length drawn from the same range, and reduces this to comparing $\Gamma(P_k)$, which converges, with $\mathbb{E}[\Gamma(R_k)]$. The expectation is exactly computable.

Theorem 13.1. *Let R_k be a uniformly random k -subset of the N odd integers in $[3, p_k]$, where $N = (p_k - 3)/2 + 1$, and put $c = (N + 1)/(k + 1)$. Then*

$$\mathbb{E}[\Gamma(R_k)] = (1 - 2^{-k}) + 2c(2 - (k + 2)2^{-k}).$$

In particular $\mathbb{E}[\Gamma(R_k)] = 1 + 4c + O(k2^{-k})$, and since $c \sim p_k/(2k)$, this diverges as soon as $p_k/k \rightarrow \infty$.

Proof. Identify the odd integers in $[3, p_k]$ with $\{1, \dots, N\}$ via $a = 2i + 1$. For a uniform k -subset the order statistics satisfy $\mathbb{E}[X_{(j)}] = j(N + 1)/(k + 1)$ exactly, so $\mathbb{E}[a_{(j)}] = 1 + 2jc$. Hence $\mathbb{E}[\Gamma] = \sum_{j=1}^k (1 + 2jc)2^{-j} = (1 - 2^{-k}) + 2c(2 - (k + 2)2^{-k})$, using $\sum_{j \leq k} j2^{-j} = 2 - (k + 2)2^{-k}$. $\square$

Remark 13.2. No prime number theorem is needed: only $p_k/k \rightarrow \infty$, i.e. that the primes have density zero, which is elementary. Monte-Carlo agreement with Theorem 13.1 is within one standard error at $k = 100, 400, 800$ ($z = -1.46, -0.38, +1.07$ over 20,000 samples).

For a statement in probability rather than in mean one needs a lower tail bound, and the following suffices.

Proposition 13.3. *Let R_k be a uniform k -subset of $\{1, \dots, N\}$ and let $X_{(J)}$ denote its J -th order statistic. For every integer $J \geq 1$ and every $\delta > 0$,*

$$\mathbb{P}[X_{(J)} \leq \delta N/k] \leq \frac{1}{\sqrt{2\pi J}} \left(\frac{e\delta}{J}\right)^J.$$

Proof. Write $m = \lfloor \delta N/k \rfloor$. The event $X_{(J)} \leq \delta N/k$ says that at least J of the k sampled points lie in $\{1, \dots, m\}$. Bounding the probability of that event by the union over all J -subsets of $\{1, \dots, m\}$ of the event that the whole subset is sampled,

$$\mathbb{P}[X_{(J)} \leq \delta N/k] \leq \binom{m}{J} \frac{k(k-1)\cdots(k-J+1)}{N(N-1)\cdots(N-J+1)} \leq \binom{m}{J} \left(\frac{k}{N}\right)^J,$$

the last step because $(k-i)/(N-i) \leq k/N$ for $0 \leq i < J \leq k \leq N$. Now $\binom{m}{J} \leq m^J/J!$ and, by Stirling, $J! \geq \sqrt{2\pi J}(J/e)^J$, so

$$\mathbb{P}[X_{(J)} \leq \delta N/k] \leq \frac{1}{\sqrt{2\pi J}} \left(\frac{emk}{JN}\right)^J \leq \frac{1}{\sqrt{2\pi J}} \left(\frac{e\delta}{J}\right)^J,$$

using $m \leq \delta N/k$. $\square$

Corollary 13.4. $\Gamma(P_k)/\Gamma(R_k) \rightarrow 0$ in probability.

Sketch. On the complement of the event above, $\Gamma(R_k) \geq 2^{-J} a_{(J)} \geq 2^{-J} (2\delta N/k)$. Fix $\delta = 1/2$ and let $J \rightarrow \infty$ slowly; the failure probability $(e/2J)^J / \sqrt{2\pi J}$ tends to 0, while $2^{-J} N/k \rightarrow \infty$ because $N/k \asymp \log k$ and J may be taken as slowly growing as one likes. Since $\Gamma(P_k)$ is bounded, the claim follows. The quantifier bookkeeping is routine and is the only part not written out here. $\square$

Remark 13.5. Proposition 13.3 was checked over 48 combinations $k \in \{100, 400, 1600, 3200\}$, $J \in \{1, 2, 4, 8\}$, $\delta \in \{0.25, 0.5, 1\}$ with 10^4 samples each; it held in every case. Numerically the smaller quantity $(e\delta/2J)^J$ also dominated the observed frequencies throughout, but this reflects only the slack in the bound and not a sharper theorem: the expected number of sample points below $\delta N/k$ is δ , not $\delta/2$, so no factor 2 is available at this step. The stated form is the one the argument proves, and the conclusion is unaffected in any case, since J is chosen after δ .

Data availability

Every numerical value quoted in this paper was produced by a script in the accompanying repository, and each script writes a log file of the same name. The scripts, the logs and the Lean development are available at <https://github.com/mathlab0911/arithmetic-landscape>.

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